

Cassandra: From Governance Infrastructure to Evidence Infrastructure

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2026-05-27

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v0.1 preprint candidate, 2026-05-27

Abstract

Public-key infrastructure is often described as a cryptographic substrate, but in practice it is also governance infrastructure: a system of certificates, policies, revocation channels, trusted lists, audits, supervisory conventions, software distribution, and institutional delegation. This paper presents Cassandra, a bounded research observatory for European list-of-lists-derived trusted-list artifacts. Cassandra records dated attempts to collect public trusted-list documents, normalizes comparable XML artifacts, emits structural diff classes, packages each substantive run into evidence bundles, exposes aggregate public dashboard data, and attaches EATF/AEP receipt metadata to the resulting evidence packages. The case demonstrates a shift from governance infrastructure to evidence infrastructure: public legal-technical artifacts can be observed, hashed, bundled, signed, displayed, and independently checked while preserving the distinction between evidence integrity and legal interpretation. Across four dated local/public lineages, Cassandra records 43 pointer attempts per run, 31 comparable normalized XML artifacts per run, 66 aggregate structural diff entries under the configured method, four EATF-verified package receipts, and five public dashboard cards. Synthetic fixtures cover stable/no-change behavior, normalized-hash changes, provider/service inventory changes, fetch failures, non-XML handling, EATF success and tamper cases, missing signing inputs, dashboard states, and claim-safety wording. Cassandra is not a trusted-list validator, not a relying-party signature validator, and not a public alerting system. Cassandra is not a supervisory tool and not a legal status oracle. Its contribution is methodological: it shows how a public governance surface can become a reproducible evidence stream, and how the same discipline can inform emerging AI governance records without collapsing cryptographic evidence into legal truth.

Keywords: evidence infrastructure; public-key infrastructure; trusted lists; eIDAS; EATF; digital government; computational public administration; reproducibility; AI governance.

Claim-Safety Note

All observations in this preprint are limited to locally recorded workflow telemetry and saved public-source artifacts: fetch metadata, parser outcomes, normalized XML hashes, structural diff classes, aggregate tables, evidence-bundle manifests, EATF/AEP receipt metadata, dashboard cards, and synthetic fixture validation outputs. A Cassandra run does not assert legal effect, does not determine whether any listed entity has gained or lost status, does not validate source signatures for relying-party purposes, does not supervise trusted lists, does not certify compliance, and does not provide public alerting. The project is an evidence-infrastructure case study, not a regulated trust service.

1. Introduction

Digital public administration increasingly depends on artifacts that are both technical and institutional. Certificates, trust lists, validation policies, audit regimes, public registers, and machine-readable administrative documents do not merely describe government infrastructure from the outside. They are part of how digital trust is governed. Yet the research record around such artifacts is often thin: a public register exists, a standard describes it, software consumes it, but the changing public state itself is not always preserved as an evidence stream that a later researcher, reviewer, or institution can replay.

Cassandra addresses this gap through a deliberately small case. It observes the European public list-of-lists and the national trusted-list documents referenced from locally saved LOTL snapshots. It records dated collection attempts, classifies endpoint and parser outcomes, normalizes comparable XML artifacts, emits configured structural diff classes, bundles each substantive run with manifests and claims, and exposes a caveated public dashboard index. The case is intentionally narrow. It does not decide legal status, validate trusted-list signatures for reliance, supervise any actor, or publish warnings. Its object is the research workflow and the saved public artifacts under that workflow.

The central claim is captured in one sentence: Cassandra moves from governance infrastructure to evidence infrastructure. PKI and eIDAS trusted lists already provide governance infrastructure: certificates, policies, supervisory lists, validation conventions, and trust-service roles. Cassandra does not replace that infrastructure. It watches one public surface of it and turns the observation process itself into evidence infrastructure: dated manifests, hashes, diffs, receipts, public cards, and explicit claim boundaries.

This matters beyond trusted lists. The EU AI Act creates adjacent evidence problems around logging, transparency, technical documentation, post-market monitoring, and lifecycle records. The AI governance problem is not only whether an AI system is safe or lawful; it is also whether claims about the system, the review, the data, the decision boundary, and the monitoring process can be bound to reproducible records. Cassandra is not an AI monitor. It is a public-governance-artifact case that makes the evidence-infrastructure problem small enough to inspect.

The paper makes four contributions. First, it gives a cautious method for longitudinal structural observation of public trusted-list artifacts. Second, it reports a working full-stack observatory with scheduled GitHub Actions, Cloudflare dashboard output, aggregate tables, figures, dashboard cards, and EATF/AEP evidence receipts. Third, it maps synthetic fixtures to reviewer-facing claims so that software behavior is tested without exposing provider or service names. Fourth, it positions Cassandra inside a thesis program in which PKI is governance infrastructure and EATF/MIRROR-style packages provide an adjacent evidence layer for AI and public-administration records.

2. Background: Trusted Lists as Governance Artifacts

The European trust-services framework is a legal-technical system, not only a family of cryptographic formats. Regulation (EU) No 910/2014 (eIDAS) establishes the EU framework for electronic identification and trust services. Regulation (EU) 2024/1183 amends eIDAS through the European Digital Identity Framework. Commission Implementing Decision (EU) 2015/1505 specifies trusted-list formats. ETSI TS 119 612 defines the trusted-list grammar; ETSI EN 319 401 describes general policy requirements for trust service providers; ETSI EN 319 411-1 and EN 319 422 connect the trusted-list layer to certificate-issuing and timestamp service governance. These sources explain why the observed XML is not arbitrary markup. It is a public administrative artifact embedded in a larger trust-services regime.

At the protocol layer, RFC 5280 anchors X.509 certificate and CRL profiles, RFC 3161 anchors timestamp protocol vocabulary, and RFC 8785 is useful for thinking about deterministic JSON evidence representations adjacent to Cassandra’s EATF/AEP package design. Outside the EU trust-list context, CA/Browser Forum baseline requirements and the PKI Consortium show that PKI governance also exists as an ecosystem of multi-stakeholder rules, assurance expectations, and operational transition problems. NIST FIPS 203, 204, and 205 further show that even core cryptographic primitives are moving under post-quantum transition pressure. The evidence layer must therefore be able to record not only stable states but also migrations, caveats, and changing assumptions.

The public-administration and infrastructure-studies literature helps name what is otherwise easy to miss. Infrastructure is relational, learned in practice, and often visible through maintenance and failure. Classifications and standards organize institutional reality; they do not merely label it. Turning-point theory helps frame why technological deployment phases demand institutional reshaping, not just invention. The useful lesson for Cassandra is modest: if public digital trust is governed through structured artifacts, then those artifacts and their changes deserve reproducible evidence records.

This background also defines the boundary. Citing eIDAS, ETSI, RFCs, NIST, CA/Browser Forum, or PKIC does not imply that Cassandra is endorsed by those bodies, implements their standards as a relying-party validator, or determines legal status. The sources establish the environment and the vocabulary. Cassandra contributes a research method for observing a public surface of that environment.

3. Related Work and Positioning

Cassandra intersects five fields.

First, PKI and cryptographic governance explain the substrate. PKI is often introduced through keys and algorithms, but it operates through certificate policies, revocation mechanisms, audits, root distribution, software defaults, identity-proofing conventions, and institutional delegation. That is governance work with cryptographic parts.¹ Cassandra uses this point to avoid treating trusted lists as mere input files. They are public records of a governance system.

Second, eIDAS and ETSI trust-service sources explain the legal-technical grammar. Trusted lists have prescribed roles and formats. This makes them attractive for struc-

¹The common shorthand “PKI equals cryptography” is useful in a classroom and dangerous in governance. The keys do not audit themselves, revoke themselves, or explain themselves to a relying party at 03:00.

tural observation, but it also makes careless claims dangerous. A structural diff is not a legal event; an endpoint failure is not a supervisory conclusion; an XML signature-shaped element is not a signature-validation result. Cassandra’s claim boundary is therefore part of the method, not a disclaimer pasted on at the end.

Third, digital-government and public-administration scholarship explain why public machine-readable artifacts matter institutionally. Work on e-Estonia, public-sector innovation, public value, trust in digital government, and AI in government all points toward the same practical problem: administrative capacity is increasingly expressed through digital artifacts that need to be maintained, interpreted, audited, and remembered.

Fourth, infrastructure studies and classification studies provide the conceptual bridge. Star and Ruhleder’s infrastructure lens, Bowker and Star’s work on classification, and related knowledge-infrastructure work all resist the fantasy that technical records are neutral pipes. They are maintained systems with categories, access conditions, and failure modes. Cassandra’s contribution is intentionally operational: it makes a specialized governance surface inspectable through manifests, hashes, diffs, dashboard cards, and receipts.²

Fifth, computational social science and digital research methods supply the cautionary frame. Repeated observation of public artifacts can produce useful evidence, but measurement validity, corpus boundaries, reproducibility, aggregation choices, and public interpretation risks must be explicit. Cassandra’s synthetic fixtures, aggregate-only prose, claim-safety checks, and abstention rules are therefore methodological features. The highest compliment for an evidence pipeline may be that it becomes boring in a reproducible way.³

4. System Overview

Cassandra is implemented as a research-only workflow around dated runs. Each run begins from a saved copy of the public European list-of-lists. The workflow extracts trusted-list pointer URLs from that saved LOTL copy, fetches the referenced public artifacts, records per-endpoint metadata, normalizes XML-like inputs where possible, records non-XML and parser outcomes, compares normalized records against a configured baseline, emits structural diff classes, builds aggregate result tables and figures, creates evidence bundles, signs or packages the run through the EATF/AEP lane when signing inputs are available, and exposes a public dashboard index.

The core files and steps are:

- `fetch.py`: collection and LOTL pointer extraction with endpoint error metadata.
- `parse.py`: deterministic XML normalization and structural extraction.
- `diff.py`: configured structural diff classes over normalized records.
- `run_daily.py`: dated orchestration with overwrite guards.
- `create_bundle.py`: MIRROR-style evidence-bundle creation.
- `scripts/eatf_package_snapshot.py`: EATF/AEP packaging integration.
- `scripts/build_observatory_index.py`: public index and dashboard-card generation.

The public observatory is deployed at <https://cassandra-observatory.pages.dev/>. Its

²Grand theory earns its keep here only if it can survive the manifest. Cassandra’s theory chapter and its hash table should be on speaking terms.

³“Boring” is not a complaint in evidence infrastructure. A reproducible system should be allowed at least one unglamorous virtue.

machine-readable index is available at <https://cassandra-observatory.pages.dev/data/index.json>. As of the 2026-05-27 build, the public index reports four runs, latest date 2026-05-27, four EATF-verified receipts, and five dashboard cards. The cards cover the latest run, EATF receipt boundary, aggregate diff classes, claim boundary, and dashboard caveat. Each card repeats that the data is structural-observation telemetry only, not trusted-list validation, signature validation, legal-status determination, supervision, compliance judgment, relying-party processing, public alerting, or publication approval.

The workflow's most important design decision is abstention. Cassandra keeps endpoint failures, non-XML artifacts, parser errors, and zero-diff days inside the record. It does not clean them away to tell a simpler story. A failed fetch is collection telemetry. A non-XML pointer is corpus-shape telemetry. A parser error is a local parser event. A zero diff means no configured structural diff was emitted by this parser/baseline pair. None of these observations determines legal status.

5. Evidence Package Design

Each substantive run can be wrapped as an evidence package. The package contains a summary artifact, manifest, claims file, notes, source copies or source references, hashes, and verification output. This is a MIRROR-style pattern: claims are made explicit, source links are recorded, and local verification is possible without asking the reader to trust a prose paragraph.

EATF/AEP adds package-level attestation semantics. For indexed ok runs, the EATF receipt verifies the corresponding package envelope, payload bytes, and declared hashes. It does not validate the underlying trusted-list source signatures, does not determine legal effect, and does not confer supervisory meaning. The distinction is crucial: evidence integrity is not legal truth. A package receipt can say that the evidence package is internally consistent under the declared verification process. It cannot say that a trust service is lawful, that a provider's status changed, or that a public authority has taken a legally significant act.

The 2026-05-27 public card records an EATF package path `evidence/2026-05-27/cassandra-observation.aep`, package hash `3ef7f33b391afac388ce4bcf9716ac7018508ed613933f4ee86`, receipt path `evidence/2026-05-27/eatf-verification.json`, receipt hash `1e971e12a46b0d8be72635671b4` and status ok. The interpretation attached to that card is deliberately narrow: ok verifies package bytes, envelope structure, and declared hashes only.

This boundary is especially important for AI governance. If evidence packages are to support AI lifecycle records, they must avoid the same overreach. A signed package can preserve what was reviewed, logged, or claimed. It does not prove that a model is safe, fair, lawful, or accurate. Cassandra provides a concrete public-artifact case for this discipline before the argument is transferred to more sensitive AI contexts.

6. Dataset and Current Results

The current Cassandra evidence base contains four dated runs: 2026-05-20, 2026-05-21, 2026-05-22, and 2026-05-27. Each run starts from 43 LOTL-derived pointer attempts. The aggregate-results table records:

Date	Pointer at- tempts	Fetch content files	Fetch errors	Normalized XML ar- tifacts	Parser errors	Diff count	Provider count total	Service count total
2026-05-20	43	41	2	31	1	0	393	4743
2026-05-21	43	42	1	31	1	0	393	4743
2026-05-22	43	42	1	31	1	26	393	4753
2026-05-27	43	42	1	31	1	40	393	4847

Across the four runs, Cassandra records 66 aggregate structural diff entries under the configured method. The dashboard card reports the class totals:

- normalized_hash_changed: 7.
- summary_field_changed: 28.
- service_inventory_changed: 5.
- provider_service_detail_changed: 26.
- listed_document_added, listed_document_removed, and provider_inventory_changed: 0.

These are structural-observation classes, not legal-effect classes. Provider and service totals are aggregate parser outputs over the saved local corpus. The paper intentionally does not name providers, services, schemes, or authorities in narrative prose. Hashed provider/service handles support local comparison while reducing the temptation to turn a research draft into a provider-specific status feed.

The results show that the workflow can preserve multiple kinds of evidence at once: collection conditions, parser boundaries, comparable records, structural changes, package hashes, EATF receipt status, public dashboard summaries, and caveats. The result is not a claim that the trusted-list ecosystem was stable or unstable. It is a claim that Cassandra can produce reproducible structural-observation telemetry over saved public artifacts and preserve enough context for later review.

7. Fixture-Backed Behavior

Real public observations are not enough to make the case publishable. The paper also needs controlled behavior tests. Cassandra therefore maintains a fixture-to-claim map that links synthetic test classes to paper claims, reviewer questions, evidence artifacts, and explicit non-claims.

The current fixture set covers:

- Stable no-change behavior: identical comparable records emit no structural diff.
- Normalized hash changes: stable record counts with changed canonical bytes are surfaced.
- Provider inventory and service inventory movements: synthetic hashed handles support aggregate comparison without real names.
- Provider-service detail changes: detail movement can be separated from whole-provider or whole-service movements.

- Fetch failures: timeout, HTTP-error, redirect, and unreachable-pointer telemetry are preserved.
- Non-XML handling: PDF, HTML, TXT, and non-parseable inputs become skips or parser events rather than fatal workflow failures.
- EATF success: a synthetic package can produce an ok receipt.
- EATF tamper: altered synthetic package bytes fail verification.
- Missing signing input: absent signing material is recorded explicitly as `skipped_missing_signing_inputs`.
- Dashboard multistate: public-index consumers can represent ok, `verify_failed`, and skipped states without turning them into alerts.
- Claim safety: configured overclaiming phrases are detected in controlled wording samples.

The fixture boundary is explicit: fixtures prove expected software behavior on synthetic inputs. They are not empirical evidence about real trusted-list content, source signatures, legal status, supervision, compliance, public alerts, or publication approval. Their role is to answer reviewer questions about the method: does the system fail closed under tampering? Does it preserve failure states? Does it avoid silent success when signing inputs are missing? Does it catch common wording overclaims? These are software and method claims, not legal claims.

8. Discussion: From PKI Governance to AI Evidence

Cassandra’s strongest thesis value is not that trusted lists are an especially fashionable dataset. It is that trusted lists sit at a rare intersection: public authority, structured technical artifacts, cryptographic governance, institutional trust, and machine-readable state. This makes them a useful case for observing how governance infrastructure can be transformed into evidence infrastructure.

The phrase “PKI as governance infrastructure” matters because it stops the analysis from shrinking PKI into algorithms. PKI includes keys and signatures, but it also includes policies, identity checks, root stores, certificate profiles, revocation conventions, timestamps, audit frameworks, trust lists, and relying-party software. Cassandra watches a public administrative surface of that infrastructure. The watched surface is not the whole governance system, but it is enough to demonstrate the evidence move.

The AI governance connection is vertical and adjacent. The AI Act does not ask Cassandra to monitor AI systems. Instead, it makes visible a similar institutional problem: records about systems, reviews, logs, risks, datasets, human oversight, transparency, and post-market monitoring need to be reproducible enough to support accountability. EATF/AEP and MIRROR-style packages offer one grammar for such records: claims, manifests, hashes, receipts, caveats, source references, and local verification. Cassandra supplies a public, non-sensitive case where that grammar can be tested against real public artifacts and synthetic failure modes.

This is also why the dashboard matters. The dashboard is not a marketing surface. It is a method surface. It exposes what the evidence stream contains and repeats what it refuses to claim. The public cards make the claim boundary visible at the point of consumption: run counts, receipt status, aggregate diff classes, and caveats travel together. That design choice is small, but it is a discipline AI governance systems will need: evidence must be packaged with its limits, not separated from them.

9. Limitations

The limitations are not incidental. They define the research object.

First, Cassandra observes saved public artifacts under one workflow. It does not establish endpoint completeness, current public availability, or authoritative state beyond the local evidence collected at a given time.

Second, the parser is not a trusted-list relying-party implementation. It records structural fields, signature-shaped elements, hashes, and parser outcomes. It does not perform source-signature validation, certificate-path validation, OCSP/CRL checks, or legal interpretation.

Third, the diff classes are descriptive buckets. A normalized-hash change, summary-field change, service-inventory movement, or provider-service-detail movement is a local structural observation. It is not a status change, compliance finding, risk score, or supervisory signal.

Fourth, the empirical series is still short. Four dated runs are enough to demonstrate a full evidence loop, public dashboard, EATF receipts, and fixture-backed behavior. They are not enough to estimate long-term churn, seasonal patterns, national differences, authority behavior, or trust-service ecosystem stability.

Fifth, the public prose is aggregate-only by design. Named examples may be useful in a later paper version, but only after operator review, source-path checks, bundle context, exact wording review, and a decision that the example improves explanation without turning the paper into a status feed.

Sixth, EATF/AEP receipts verify package integrity under the declared process. They do not validate underlying legal truth. This limitation is also the point. Evidence infrastructure becomes credible when it knows where evidence stops.

10. Conclusion

Cassandra shows that a public governance artifact can be turned into a bounded evidence stream without pretending to become a regulator, validator, or oracle. The system collects public LOTL-derived artifacts, preserves endpoint and parser telemetry, normalizes comparable XML, emits structural diff classes, packages run evidence, attaches EATF/AEP receipts, exposes public dashboard cards, and validates its own claim boundaries through synthetic fixtures.

The case supports a broader thesis: PKI did not merely secure digital transactions; it institutionalized a way of governing digital trust through certificates, lists, policies, audits, revocation, timestamps, and validation conventions. AI governance now needs an adjacent evidence infrastructure for systems, decisions, reviews, logs, public artifacts, and lifecycle records. Cassandra is the public-governance-artifact case. Its lesson is deliberately restrained: evidence integrity can be made visible and reproducible, but it must not be confused with legal truth.

Artifact Availability

Repository: <https://github.com/tyche-institute/cassandra> Dashboard: <https://cassandra-observatory.pages.dev/>

Public index: <https://cassandra-observatory.pages.dev/data/index.json>

Key local artifacts for this preprint:

- paper/preprint/cassandra-preprint-v0.1.md
- paper/reference-seed-bibliography.md
- paper/claims-and-evidence-table.md
- notes/cassandra-checked-reference-ledger-2026-05-27.md
- notes/cassandra-checked-reference-url-validation-output.json
- notes/fixture-to-claim-map.md
- notes/fixture-matrix.md
- notes/case-study-maturity-matrix.md
- notes/publication-status-discipline.md
- observatory/public/data/index.json
- observatory/public/data/cards/index.json
- evidence/2026-05-27/eatf-verification.json

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